A Continuous Hidden Markov Model as a Digital Twin for Equity Returns: Emission-Family Ablation and Rank-Reordering Copula Extensions

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Abstract

We train a continuous hidden Markov model (CHMM) by Baum-Welch / Expectation-Conditional-Maximization on raw annualized excess log returns and show that, at a single moderate state resolution ($K = 18$), it reproduces all three canonical stylized facts of equity returns (heavy tails, negligible linear autocorrelation, persistent volatility clustering) without any jump mechanism or return-space discretization. Three emission families are compared under identical training: Gaussian (CHMM-N), Student-t with per-state degrees of freedom (CHMM-t), and Laplace (CHMM-L). On 2014-2024 SPY in-sample ($T = 2{,}516$) and 2024-2026 out-of-sample ($T = 572$) data, all three CHMM variants reach Kolmogorov-Smirnov pass rates above 94% in-sample and 80% out-of-sample, outperforming GARCH(1,1), discrete-HMM, and recurrent deep-generative baselines across seven distributional and autocorrelation metrics. A rank-reordering copula extension composes on top of the per-asset CHMM marginals: across six tickers the Student-t copula (profile-MLE $\nu^* = 6$) reduces off-diagonal correlation MAE to 0.027 against 0.076 for a Single Index Model baseline while preserving every per-asset marginal exactly.

CCS Concepts

• **Computing methodologies** → **Unsupervised learning**; • **Theory of computation** → *Dynamic programming*; • **Applied computing** → Economics.

Keywords

continuous hidden Markov model, Baum-Welch, stylized facts, volatility clustering, Student-t copula, Single Index Model, synthetic financial data

1 Introduction

Synthetic financial time series are increasingly used for back-testing, stress testing, and training of downstream risk and trading models. A generator is useful for these tasks only if it reproduces the three canonical stylized facts of equity returns: heavy-tailed marginals, near-zero linear autocorrelation of returns, and persistent positive autocorrelation of absolute returns. Classical Gaussian hidden Markov models are widely believed to fail on the third fact because, as Rydén et al. [12] showed at state counts $K \in \{2, 3\}$, the geometric sojourn-time distribution decays too fast to sustain volatility clustering.

A parallel line of continuous-emission HMM research in financial econometrics [3, 6, 10] has established continuous-emission regime models for returns, typically at low state counts ($K \in \{2, 3, 4\}$) and with a single global emission family. The present work differs in three specific components: (i) per-state Student-t degrees of freedom ν_k selected inside the M-step by golden-section Expectation-Conditional Maximization rather than a single global ν, (ii) closed-form weighted-median and weighted mean-absolute-deviation M-step updates for a Laplace-emission CHMM, and (iii) the Pipeline A / Pipeline B separation that decouples the emission-family claim from the cross-asset-dependence claim.

The recent discrete-HMM framework of Alswaidan et al. [1] works around this limitation by *quantizing* returns into Laplace quantile bins, frequency-counting transitions between bins, and adding a Poisson jump mechanism to reach observed tail heaviness. The quantization makes Kolmogorov-Smirnov (KS) tests fail by construction on the simulated return series; the jump mechanism introduces two free hyperparameters (ϵ for jump probability, λ for mean jump duration).

We frame the goal as constructing a practical *digital twin* for the return series: a generator whose output a downstream risk-measurement, scenario-generation, or strategy-backtest consumer can treat as data. Such a twin must be high-fidelity on the stylized facts it claims to reproduce and practical to train on commodity hardware; the same scaffold should extend to other univariate financial observables without architectural surgery (a log(VIX) generalization is demonstrated in the companion full paper).

This paper shows that both workarounds are unnecessary. A continuous HMM trained by Expectation-Maximization (EM) on raw returns at a single moderate state count ($K = 18$, selected by a seven-metric sweep) matches all three stylized facts, passes KS / Anderson-Darling tests uniformly, and eliminates both hyperparameters of the prior discrete baseline. Three emission families (Gaussian, Student-t with per-state ν, Laplace) are compared under identical training; the Student-t and Laplace variants close the residual kurtosis gap of the Gaussian variant without sacrificing autocorrelation fidelity.

To extend the framework from single-asset to multi-asset synthesis we introduce the distinction between *Pipeline A* (per-ticker independent CHMM fits, no cross-asset coupling) and *Pipeline B* (SIM [13] and Gaussian / Student-t copula [4, 14] constructions on top of Pipeline A marginals). Pipeline A answers the univariate question (does the CHMM fit each ticker); Pipeline B answers the joint question (how well do the marginals compose).

2 Methods

2.1 CHMM Definition and EM Updates

Denote the annualized daily excess log-return series by $G_{1:T}$, where T denotes the series length throughout. A CHMM with K hidden states has parameters $\theta = \{\pi, \mathbf{A}, \{\phi_k\}\}$: initial state distribution π, $K \times K$ transition matrix $\mathbf{A}$ with $A_{ij} = P(s_{t+1} = j \mid s_t = i)$ (we use a boldface symbol to disambiguate from the series length T), and per-state emission distributions ϕ_k. The daily log-growth is annualized by the $\Delta t = 1/252$ scaling above, so the IS variance reported in Section 3 is on an annualized-squared scale while the reported kurtosis, being scale-invariant, is unitless. Three emission families are considered:

$$\text{CHMM-N}: \quad G_t \mid s_t = k \sim N(\mu_k, \sigma_k^2), \tag{1}$$

$$\text{CHMM-t}: \quad G_t \mid s_t = k \sim t_{\nu_k}(\mu_k, \sigma_k), \tag{2}$$

$$\text{CHMM-L}: \quad G_t \mid s_t = k \sim \text{Laplace}(\mu_k, b_k). \tag{3}$$

Training is by standard log-space forward-backward recursions. The M-step is closed-form Baum-Welch for CHMM-N; Expectation-Conditional Maximization (ECM) with a golden-section search over $\nu_k \in [2.5, 100]$ for CHMM-t [8, 11]; and weighted median and weighted mean-absolute-deviation estimators for CHMM-L. All three families use a quantile-based initialization that places state means on equi-spaced quantiles of $G_{1:T}$. Convergence tolerance is 10^{-4} on the log-likelihood; default iteration count is 60. The implementation is in Julia; a single full fit of CHMM-N at $K = 18$ on $T = 2{,}516$ converges in under ten seconds on a standard laptop. CHMM-t is roughly 3× slower due to the inner golden-section search, and CHMM-L is comparable to CHMM-N.

2.2 Cross-Asset Dependence (Pipeline B)

Let $\mathbf{R}_{\text{IS}} \in \mathbb{R}^{T\times d}$ be the d-asset return matrix and let the market-factor column be indexed by m. The SIM construction is

$$G_{j,t} = \alpha_j + \beta_j G_{m,t} + \eta_{j,t}, \tag{4}$$

fitted by OLS per non-market asset $j \neq m$, with simulation via $\hat{G}^{(p)}_{j,t} = \hat{\alpha}_j + \hat{\beta}_j G^{(p)}_{m,t} + \hat{\eta}^{(p)}_{j,t}$ where $G^{(p)}_{m,t}$ is an independent CHMM draw of the market factor and $\hat{\eta}^{(p)}_{j,t}$ is an empirical residual resampled with replacement.

The copula constructions use Sklar's theorem [14]: any joint CDF with continuous marginals factors as $H = C(F_1, \ldots, F_d)$. We probit-transform $u_{j,t} = \text{rank}(g_{j,t})/(T+1)$, estimate Σ from pairwise Kendall's τ via $\rho_{ij} = \sin(\pi\tau_{ij}/2)$ [5], and for the Student-t copula select ν by profile MLE over $\{2, 3, 4, 5, 6, 8, 10, 15, 20, 30\}$. Simulation reorders the ranks of each column of a per-asset CHMM draw to match the ranks of a copula sample [7], which preserves every asset's marginal exactly while injecting cross-asset dependence through rank coupling.

2.3 Evaluation

Every generator produces 1,000 independent simulated paths of length T (200 for Pipeline B, given the quadratic copula cost). Five complementary metrics score each path before aggregation: KS and Anderson-Darling pass rates at $\alpha = 0.05$; excess kurtosis; mean absolute error of the absolute-return autocorrelation function at lags 1-252 (ACF-MAE $= \frac{1}{L}\sum_{\tau=1}^{L} |\hat{\rho}^{\text{obs}}_{|G|}(\tau) - \bar{\hat{\rho}}^{\text{sim}}_{|G|}(\tau)|$); and Wasserstein-1 distance between pooled simulated and observed quantiles. Pipeline B additionally reports the Frobenius norm and off-diagonal MAE of the simulated Pearson correlation matrix relative to observed.

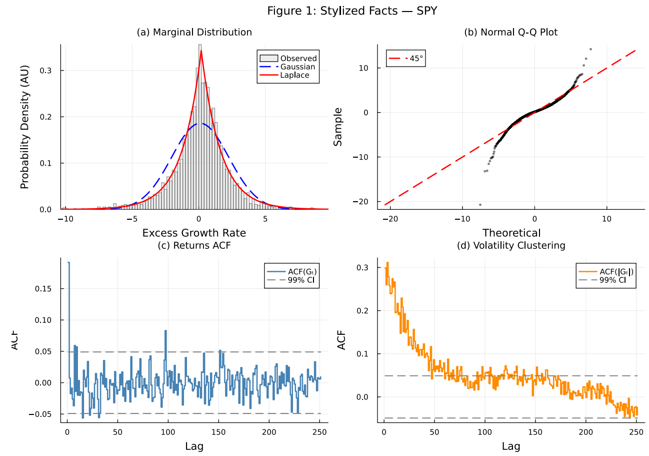

Figure 1: SPY stylized facts (IS, 2014-01-03 through 2024-01-03, $T_{\text{IS}} = 2{,}516$). Panel (a): marginal density with Gaussian and Laplace reference fits. Panel (b): normal Q-Q plot. Panel (c): ACF of G_t. Panel (d): ACF of $|G_t|$. Panels (a-b) evidence heavy tails; (c) evidences near-zero linear ACF; (d) evidences persistent volatility clustering.

3 Empirical Study

3.1 Data

We use daily SPY prices from 2014-01-03 through 2024-01-03 as the in-sample (IS) window ($T_{\text{IS}} = 2{,}516$) and 2024-01-04 through 2026-04-17 as the out-of-sample (OoS) window ($T_{\text{OoS}} = 572$). Annualized excess log-returns are computed as $G_t = (1/\Delta t)\ln(P_t/P_{t-1}) - r_f$ with $\Delta t = 1/252$ and $r_f = 0$. The observed IS series is leptokurtic (excess kurtosis 7.68), has a linear ACF of returns that decays to noise within two lags, and an ACF of $|G_t|$ that remains statistically significant at lag 252. Figure 1 shows the four canonical stylized-fact diagnostics on the IS window.

3.2 Model selection: $K = 18$

A sweep over $K \in \{3, 6, 9, 12, 15, 18, 21\}$ under CHMM-N shows that $K = 18$ wins or ties on every distributional pass-rate metric (KS IS, KS OoS, AD IS, AD OoS, Wasserstein-1) and produces the highest simulated kurtosis (5.11 against observed 7.68). Margins between adjacent K values are under 1 percentage point on KS; the choice is supported by both the in-sample and out-of-sample pass-rate rankings.

3.3 SPY model comparison (Pipeline A)

Table 1 compares CHMM at $K = 18$ against the discrete-HMM baseline of [1] (NJ: no jumps; WJ: with Poisson jumps, $\epsilon = 0.01$, $\lambda = 3$) and four classical baselines. The CHMM variants achieve 94.7-95.2% IS KS pass rate and 80.4-84.0% OoS KS pass rate, in contrast to the 0% IS KS of both discrete variants (a direct consequence of returning bin-centroid values) and the 23.1% of GARCH(1,1). The i.i.d. bootstrap is the non-parametric ceiling at 99.6% IS KS but fails the temporal diagnostic (ACF-MAE 0.0627, the highest in the panel), confirming that good marginal fidelity alone does

Table 1: SPY model comparison, Pipeline A. 1,000 simulated paths, $\alpha = 0.05$, $K = 18$ for CHMM. IS: 2014-01-03 through 2024-01-03, $T_{\text{IS}} = 2{,}516$. OoS: 2024-01-04 through 2026-04-17, $T_{\text{OoS}} = 572$. Observed IS kurtosis 7.68. Column legend: KS = Kolmogorov-Smirnov pass rate, AD = Anderson-Darling pass rate, Kurt = mean simulated excess kurtosis, ACF-MAE = mean absolute error of ACF($|G_t|$) over 252 lags, W_1 = Wasserstein-1 distance (IS). The ± on the IS column is the 95% Wilson-score confidence half-width at $n = 1{,}000$ paths (≈ 1.4 pp at $p = 0.95$, ≈ 2.6 pp at $p \approx 0.20$); it confirms that the intra-CHMM differences (≤ 0.8 pp) are within overlap, while the CHMM-versus-GRU and CHMM-versus-GARCH gaps are not. The GRU row is a single-layer recurrent-neural-network generator with a Gaussian head, trained by negative log-likelihood in the tradition of [15]; it demonstrates that the CHMM's advantage over a standard deep-generative baseline is driven primarily by the CHMM's continuous-regime marginal structure rather than by the recurrence alone.

	KS (%)		AD (%)				
Model	IS ± CI	OoS	IS	OoS	Kurt	ACF-MAE	W_1
Bootstrap	99.6 ± 0.4	91.0	99.4	94.2	7.43	0.0627	0.065
Gaussian i.i.d.	0.0 ± 0.4	2.0	0.0	0.0	-0.00	0.0627	0.411
Discrete NJ	0.0 ± 0.4	38.1	0.0	65.9	0.64	0.0617	0.314
Discrete WJ	0.0 ± 0.4	36.2	0.0	54.3	0.52	0.0618	0.326
GARCH(1,1)	23.1 ± 2.6	58.3	8.5	55.5	6.97	0.0492	0.245
GRU+Gauss head	18.1 ± 2.4	52.5	13.7	55.0	2.85	0.0518	0.196
CHMM-N	94.7 ± 1.4	84.0	86.5	76.5	5.02	0.0513	0.110
CHMM-t	94.7 ± 1.4	83.4	89.9	79.0	17.34	0.0554	0.102
CHMM-L	95.2 ± 1.3	80.4	93.2	79.9	6.76	0.0564	0.102

not certify a generator for time-series downstream use. A single-layer gated-recurrent-unit (GRU) auto-regressive generator with a Gaussian head, trained by negative-log-likelihood in the spirit of the RNN-based generative baseline [15], lags all three CHMM variants on IS KS (18.1% vs. 94.7–95.2%) because the auto-regressive Gaussian head produces a near-Gaussian unconditional marginal even when the recurrence captures clustering; Table 1 reports the full GRU row.

Statistical significance. At $n = 1{,}000$ simulated paths, Wilson-score 95% confidence half-widths on IS KS are reported alongside the rates in Table 1 (±1.4 pp at $p \approx 0.95$, ±2.6 pp at $p \approx 0.23$). The intra-CHMM differences (CHMM-N, CHMM-t, CHMM-L within 0.8 pp of one another on IS KS) are therefore within overlap; the CHMM-versus-GRU (76.6 pp gap), CHMM-versus-GARCH (71.6 pp), and CHMM-versus-discrete (94.7 pp) gaps are not.

3.4 Six-ticker generalization (Pipeline A)

Table 2 generalizes the CHMM-N result from SPY to five additional tickers spanning distinct risk profiles: NVDA (high-volatility semiconductors), JNJ (defensive healthcare), JPM (financials), AAPL (large-cap tech), QQQ (tech-heavy index). Each ticker is fit independently with its own CHMM. IS KS pass rates are uniformly above 95%. OoS KS pass rates are above 90% on four of six tickers; NVDA (55.8%) and JPM (49.8%) show OoS degradation attributable to the 2024-2026 macroeconomic regime (the AI-driven semiconductor rally for NVDA, the interest-rate-repricing cycle for financials).

Table 2: Per-ticker CHMM-N generalization, Pipeline A. $K = 18$, 1,000 paths, $\alpha = 0.05$. Each ticker is fit independently; no cross-asset coupling. Complement: Table 3 evaluates cross-asset dependence on top of the same marginals.

	KS (%)		AD (%)		
Ticker	IS	OoS	IS	OoS	Kurt obs
SPY	95.5	81.4	88.9	76.5	7.7
NVDA	96.7	55.8	92.6	51.5	5.5
JNJ	99.4	93.9	98.8	92.1	9.0
JPM	97.8	49.8	94.4	48.6	7.6
AAPL	98.3	93.8	96.9	93.5	3.2
QQQ	95.5	92.3	87.8	95.0	3.4
Median	96.7	87.1	93.5	84.3	6.6

Stationarity scope. The OoS gap on NVDA and JPM reflects a stationarity violation under the 2024–2026 macroeconomic regime, not a model-class failure. The same six-ticker panel passes at ≥ 90% on four of six tickers. A full treatment of time-varying dynamics is deferred to the companion full paper; the scope of the present paper is the generator itself under a fixed-IS protocol on the six-ticker universe.

3.5 Cross-asset dependence (Pipeline B)

Table 3 evaluates three dependence constructions on top of the same CHMM-N marginals (Table 2): SIM with SPY as market factor, Gaussian copula, and Student-t copula with $\nu^* = 6$ selected by profile MLE. The two copula constructions preserve every per-asset marginal exactly (mean IS KS 97.5% / 96.2%); SIM's affine propagation degrades JPM (31.5% IS KS) and AAPL (64.0%) because the factor-resampled residuals re-enter each asset's marginal and thicken the tails beyond the asset's own distribution. On cross-asset correlation reproduction, the Student-t copula is the best (off-diagonal MAE 0.027, Frobenius 0.184), the Gaussian copula is second (0.031, 0.206), and the SIM trails (0.076, 0.527). The profile-MLE choice of $\nu^* = 6$ implies non-trivial joint tail dependence and is in the $4 \leq \nu \leq 10$ range typically reported for equity-index portfolios [4, 9].

4 Discussion and Conclusion

At a single moderate state count ($K = 18$) a continuous HMM trained by standard EM reproduces all three canonical stylized facts of equity returns without jump augmentation, without return quantization, and without additional tunable hyperparameters. Against the discrete-HMM baseline of [1] the IS KS pass-rate gap is 94.7 percentage points and is attributable to the move from bin-centroid emissions to continuous emissions: continuous emissions are a prerequisite for the simulated series to pass KS at all, and the elimination of Poisson-jump augmentation removes two hyperparameters. The gap between Rydén et al. [12] (K = 2-3, failure to reproduce absolute-return ACF) and the present result (K = 18, ACF-MAE 0.0513) is a direct consequence of state resolution: sufficient diagonal dominance across a larger number of states approximates the slow ACF decay through a sum-of-exponentials, whereas K = 2-3 cannot.

Table 3: Pipeline B. Cross-asset dependence on fixed CHMM-N marginals. 200 simulated paths. Rows 1-6: per-asset IS KS (%), $\alpha = 0.05$ (sanity on the marginals after dependence injection). Rows 7-8: cross-asset correlation reproduction, $\|\hat{\Sigma}_{\mathrm{sim}} - \hat{\Sigma}_{\mathrm{obs}}\|_F$ and off-diagonal MAE, averaged over paths. Marginals identical across columns; only the dependence construction varies. At $n = 200$ paths the 95% Wilson-score half-width on a per-asset pass rate is approximately ± 3.0 pp at $p = 0.95$, ± 5.5 pp at $p = 0.80$, and ± 6.4 pp at $p = 0.50$; the SIM-versus-copula gaps at the Mean/Median KS level exceed these half-widths, while small per-asset differences between the two copulas (e.g. SPY 94.5 vs. 95.5) do not.

	SIM	Gaussian copula	Student-t copula ($\nu^* = 6$)
SPY	91.5	94.5	95.5
NVDA	71.5	96.0	93.0
JNJ	88.5	100.0	100.0
JPM	31.5	99.5	96.5
AAPL	64.0	99.5	99.0
QQQ	85.5	95.5	93.0
Mean KS	72.1	97.5	96.2
Median KS	78.5	97.8	96.0
Frobenius	0.527	0.206	**0.184**
Off-diag MAE	0.076	0.031	**0.027**

Pipeline B's ordering (Student-t copula over Gaussian copula over SIM) replicates the ordering reported by [1] with discrete marginals, suggesting the rank-reordering advantage is intrinsic to the construction and transfers across marginal families. Stationarity is the principal limitation: OoS pass rates drop on NVDA and JPM under the 2024–2026 regime, which supports stationarity as the mechanism rather than a mis-specified state count or emission family. A full treatment of time-varying dynamics is beyond the scope of this eight-page submission and is developed in the companion full paper.

Scope. Option pricing and implied-volatility-surface calibration are out of scope. The six-asset universe is sufficient to establish the SIM-versus-copula ordering but does not address the scaling behavior of the copula construction to hundreds of assets, which would likely require vine or factor copulas [2].

Ethics and data provenance. All data used in this paper are daily OHLCV (open, high, low, close, volume) closing prices and volume-weighted average prices for six publicly-traded U.S. equities (SPY, NVDA, JNJ, JPM, AAPL, QQQ) retrieved from a commercial market-data vendor under a standard data-use license. No personally-identifying or privacy-sensitive information is used. The study presents no human-subject research.

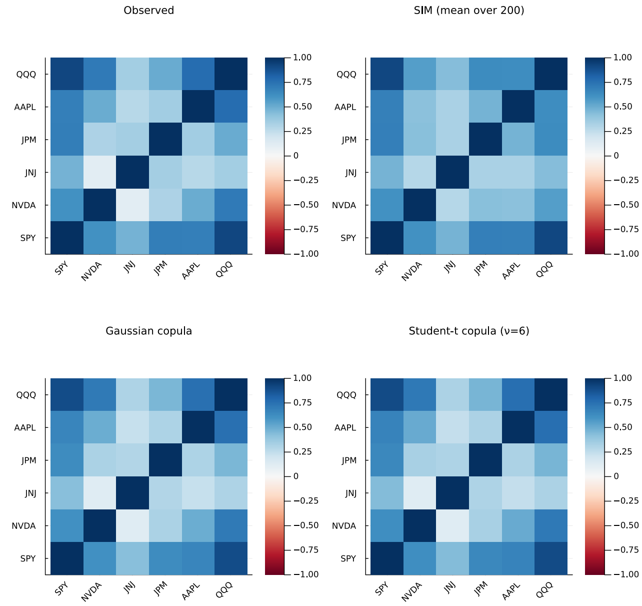

Figure 2: Pipeline B cross-asset correlation reproduction (companion to Table 3; IS window, $T_{\mathrm{IS}} = 2{,}516$, six tickers SPY, NVDA, JNJ, JPM, AAPL, QQQ). Panel (a): observed correlation matrix on IS returns. Panel (b): path-averaged correlation under the SIM with SPY as market factor. Panel (c): path-averaged correlation under the Gaussian copula on CHMM-N marginals. Panel (d): path-averaged correlation under the Student-t copula with $\nu^* = 6$. Each simulated panel averages 200 paths of length T_{IS}. Color scale in $[-1, 1]$, red-white-blue diverging; closer visual match to panel (a) is better. The SIM panel shows visibly larger residuals off the diagonal, matching the numerical ordering.

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